Plasma Sources
Science and
Technology

PAPER

Design and characterization of the Magnetized Plasma Interaction Experiment (MAGPIE): a new source for plasma–material interaction studies

To cite this article: Boyd D Blackwell *et al* 2012 *Plasma Sources Sci. Technol.* **21** 055033

IOP Publishing — Plasma Sources Science and Technology
Plasma Sources Sci. Technol. **21** (2012) 055033 (7pp) doi:10.1088/0963-0252/21/5/055033

Design and characterization of the Magnetized Plasma Interaction Experiment (MAGPIE): a new source for plasma–material interaction studies

Boyd D Blackwell, Juan Francisco Caneses, Cameron M Samuell, John Wach, John Howard and Cormac Corr

Plasma Research Laboratory, Research School of Physics and Engineering, The Australian National University, Canberra 0200, Australia

E-mail: boyd.blackwell@anu.edu.au

Received 25 March 2012, in final form 7 September 2012
Published 4 October 2012
Online at stacks.iop.org/PSST/21/055033

Abstract

The Magnetized Plasma Interaction Experiment (MAGPIE) is a versatile helicon source plasma device operating in a magnetic hill configuration designed to support a broad range of research activity and is the first stage of the Materials Diagnostic Facility at the Australian National University. Various material targets can be introduced to study a range of plasma–material interaction phenomena.

Initially, with up to 2.1 kW of RF at 13.56 MHz, argon (10^{18}–10^{19} m^{-3}) and hydrogen (up to 10^{19} m^{-3} at 20 kW) plasma with electron temperature ∼3–5 eV was produced in magnetic fields up to ∼0.19 T. For high mirror ratio we observe the formation of a bright blue core in argon above a threshold RF power of 0.8 kW. Magnetic probe measurements show a clear $m = +1$ wave field, with wavelength smaller than or comparable to the antenna length above and below this threshold, respectively. Spectroscopic studies indicate ion temperatures <1 eV, azimuthal flow speeds of ∼1 km s^{-1} and axial flow near the ion sound speed.

(Some figures may appear in colour only in the online journal)

1. Introduction

Recent reviews [1] in the EU and the USA have confirmed that the science and technology of materials under extreme heat loads are critical to the success of plasma fusion reactors such as ITER and the future DEMO, and is an area requiring significant research and development. This inspired a movement to tackle plasma–material science issues in a systematic way using specially designed linear devices [2], and more recently several variants [3]. These have the advantage compared with tokamaks that the plasma conditions can be carefully controlled and monitored for long periods of time. The properties of the plasma source can be tailored to study the phenomena of interest. The Plasma Research Laboratory of the Australian National University has recently constructed a new linear magnetized plasma device, MAGPIE: the Magnetized Plasma Interaction Experiment, employing electrodeless plasma generation by helicon wave heating. Target parameters are $T_e \sim 5$ eV, $n_i > 10^{19}$ m^{-3} *(in hydrogen)* with thermal ion flux $>2 \times 10^{22}$ m^{-2} s^{-1}, and $>10^{23}$ m^{-2} s^{-1} with bias.

Helicon waves are bounded electromagnetic waves in magnetized plasma with frequencies between the ion cyclotron frequency and the electron cyclotron frequency [4] and are of considerable interest because they can propagate at much higher densities than many other waves in plasmas. Helicon wave generated plasma discharges provide an opportunity to reach very high densities in electrodeless discharges with very high fractional ionization. Over the past three decades, helicon discharges have been used to study a number of astrophysical and fusion-related phenomena including wave physics, turbulence and the related driving mechanisms of these phenomena [5–11]. Helicon sources have become increasingly important in material processing and ion

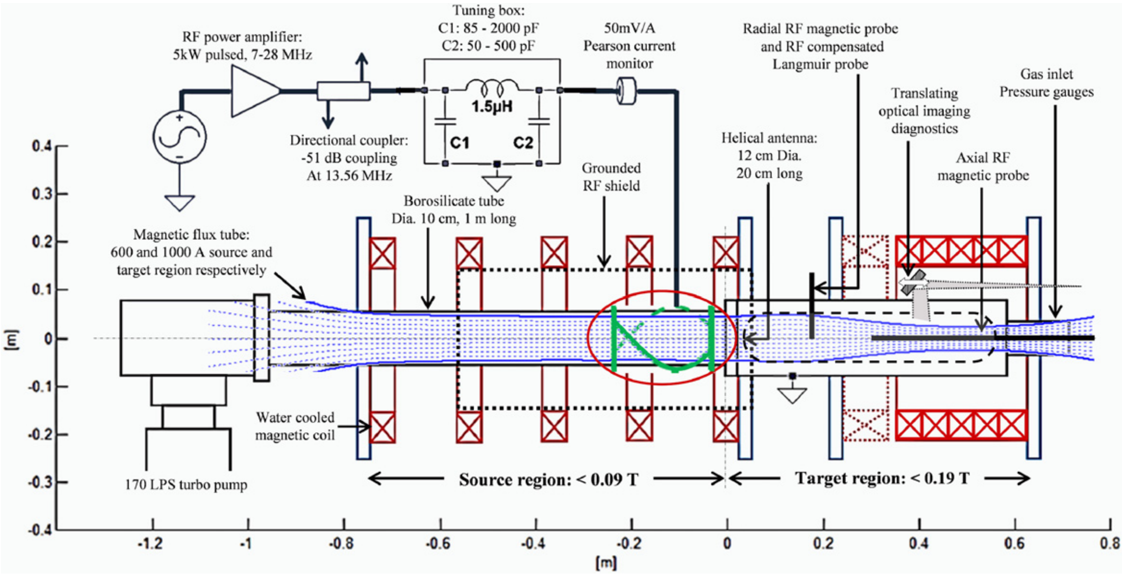

Figure 1. Schematic view of the MAGPIE showing its main components: the vacuum vessel, the RF plasma source and the magnets. The positions of the antenna, imaging system, RF magnetic and Langmuir probe are indicated.

propulsion applications [12, 13]. Such sources are capable of producing high-density plasmas ($\sim 10^{19}$ m^{-3}) in both heavy gases such as argon [14, 15] and at higher powers in light gases such as hydrogen [16, 17] and therefore provide an opportunity to reproduce conditions in the divertor region of a fusion plasma. In addition, linear plasma machines provide better diagnostic access allowing the application of sophisticated diagnostic techniques that are not easily implemented to a tokamak environment.

MAGPIE, the first stage of the Materials Diagnostic Facility, launches helicon waves into an increasing magnetic field region to create a high-density plasma downstream of the source region to study basic plasma phenomena, materials and their interaction with plasma under extreme conditions, and to serve as a test bed for developing diagnostics for the edge regions of a fusion reactor. A range of electrically isolated or biased target materials can be inserted into the higher density, compressed plasma region. Using various diagnostics, including new imaging Doppler systems [18, 19], we will study the physics of plasma generation, helicon wave propagation and damping in the inhomogeneous MAGPIE plasmas.

In this paper we describe the experimental setup of MAGPIE, namely its magnetic system, vacuum vessel and RF plasma source, and characterize the plasma device by presenting some initial measurements of ion density, electron temperature, helicon wave fields and optical emission in argon plasmas.

2. Device description

The device, which is shown schematically in figure 1, consists of two parts: a source region and a target region. The source region is a 1 m long, borosilicate tube 10 cm in diameter. The source tube is attached to an aluminium target chamber 0.7 m in length. Both the source and target chambers are positioned coaxially inside two sets of water-cooled solenoids, each of 0.3 m internal diameter. The target chamber sits in the high field region of the device.

The operating gas is fed at the target chamber end and removed at the other by a throttled 170 l s^{-1} turbo pump, which routinely provides a base pressure of approximately 10^{-4} Pa. This provides a gradient of fill pressure in the same sense as the expected gradient in plasma density, and allows better access by removing the pump from the target region. The system pressure is measured in the upstream region using ionization, Convectron and Baratron gauges. In the present configuration, two large Pyrex windows are positioned on the sides of the target chamber (dashed in figure 1) to provide optical access over a large region (80 mm $\times$ 500 mm). The target chamber has a number of fixed ports on the sides and on the end-plate for probe access.

The RF heating system currently consists of an A-150 ENI 150 W broadband (3–35 MHz) amplifier driving an Alpha 77-Dx 5 kW pulsed RF amplifier and a custom-built 20 kW pulsed amplifier. The matching network is a ‘pi’ type comprising two variable vacuum capacitors with capacitances of C1 = 85–2000 pF, 5 kV and C2 = 50–500 pF, 15 kV (replaced by 85–2000 pF for 7 MHz) and a 1.5 μH inductor. This matchbox can support a frequency range 7–28 MHz. Forward and reflected powers are measured with a directional coupler and an Alpha A4520 5 kW power meter. A 50 mV A^{-1} Pearson 6600 current monitor installed around the antenna feed-through measures the antenna current. The antenna is a 20 cm long, 12 cm in diameter Nagoya III helicon antenna with a 180° helical twist. The antenna is left-handed, implying that $m = +1$ waves are launched towards the target, antiparallel to the dc magnetic field vector [20].

The magnetic field is used to support the generation of helicon waves for plasma production, provide some degree of radial confinement, and to assist transport of the plasma to the target region. The axial gradient is intended to increase the density in the target chamber [17, 21]. In order to tailor the

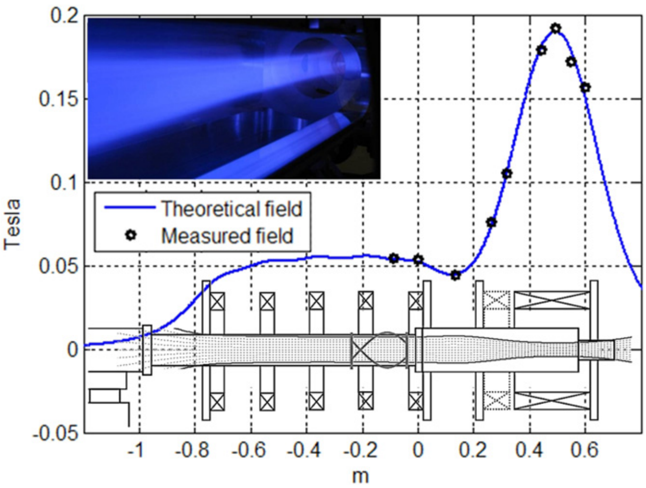

Figure 2. The maximum non-uniform axial magnetic field strength configuration along the axis with mirror ratio 3.45. Inset: magnetic field lines and image of the magnetically pinched plasma column in the target chamber.

shape of the magnetic field, there are two separate solenoid sets. One set of five solenoids is used to generate a field in the source region with a maximum strength of 0.09 T. The other set of five provides a much stronger field of up to 0.19 T in the target region. Each individual solenoid has 0.3 m internal diameter, approximately 14 turns, 3.3 mΩ of resistance and covered with an epoxy resin. The two sets of solenoids are water-cooled and powered separately by two 1000 A, 20 V dc power supplies. Varying the relative current in these two power supplies allows for the mirror ratio to be controlled. By selecting which of the target region solenoids are powered (indicated with solid lines in figure 1), the axial position of the magnetic hill can be varied.

Figure 2 shows the calculated and measured non-uniform axial magnetic field of MAGPIE with maximum field in the target region, and the insets show the compression of the flux tubes and the resulting focusing effect on the plasma in the target chamber.

3. Diagnostics

During this initial stage we used Langmuir probes to measure plasma density and electron temperature profiles in the target region. Additionally, we used RF magnetic probes (‘B dot probes’) to measure the helicon wave field structures downstream of the antenna. Optical spectroscopic ‘coherence imaging’ was deployed to measure the spatial variation of the velocity distribution function of Ar II ions during the formation of blue-core plasmas.

3.1. RF compensated Langmuir probe

A passively compensated Langmuir probe [22] was used to find the plasma parameters from measurements of the probe $I(V)$ characteristics using an ‘Impedans’ data acquisition system. The ion densities were calculated from the saturation currents as determined from the $I(V)$ curve [23]. The compensated Langmuir probe design consists of a 0.1 mm diameter platinum wire with 6 mm extending beyond the surrounding alumina insulator to form the probe tip. The probe was cleaned at regular intervals in argon discharges by drawing electron currents. The probe was positioned 17 cm downstream of the source/diffusion chamber interface (refer to figure 1).

3.2. Symmetric double Langmuir probe

A symmetric double probe is employed to measure ion densities and electron temperatures during high-power pulsed discharges (>10 kW). The probe consists of two molybdenum tips 5 mm in length and 0.1 mm in diameter, separated by a distance of 3 mm and oriented perpendicular to the local magnetic field. The probe axially samples the plasma from under the antenna down to the peak of the magnetic field in the target region. A floating differential voltage is swept across the probe tips at 17 kHz using a transformer isolated amplifier system. The probe voltage and the plasma current are measured with a system of high-voltage isolating differential amplifiers. Ion saturation currents and electron temperatures are calculated as outlined in Swift and Schwar [23].

As a double probe samples the energetic electron population [23] this may lead to an overestimation of the electron temperature and thus an underestimation of the ion density. The smaller currents reduce plasma perturbation and probe thermal damage under high-power operation. In addition, double probes have a simpler and more rugged construction than RF compensated probes while retaining effective rejection of RF potential fluctuations [24, 25].

3.3. RF magnetic probe

The RF magnetic probe used in this experiment consists of two sensing coils oriented radially and axially with respect to the longitudinal axis of the probe shaft. Each coil is 3.5 mm in diameter, and has 6 turns of 0.5 mm copper wire.

The probe shaft is a grounded 1.7 m long stainless steel shaft, which is inserted into a borosilicate tube 1.7 m in length and 12 mm in outer diameter. The length of the probe allows measurement of RF wave fields from under the antenna down to the end of the target region. Each coil is connected to a 22 turn 1 : 1 transformer used as a current balun for common mode rejection using two parallel 50 Ω coaxial cables of 1.8 m in length. The probe is capable of sampling two mutually perpendicular magnetic field components simultaneously.

The RF magnetic probe couples inductively to the magnetic components of the helicon wave but also couples electrostatically to the RF time varying plasma potential of the plasma. The current balun exhibits a very high impedance for common mode signals (electrostatic) and low impedance for differential signals (magnetic), thus rejecting the electrostatic component relative to the magnetic component that corresponds to the helicon wave. Figure 3 shows a schematic of the RF magnetic probe. The probe was calibrated using a 2 turn, 25 cm diameter Helmholtz coil from 3 to 30 MHz. The Helmholtz coil current was calculated by measuring the voltage drop across the coil and allowing for its measured inductance and parasitic capacitance.

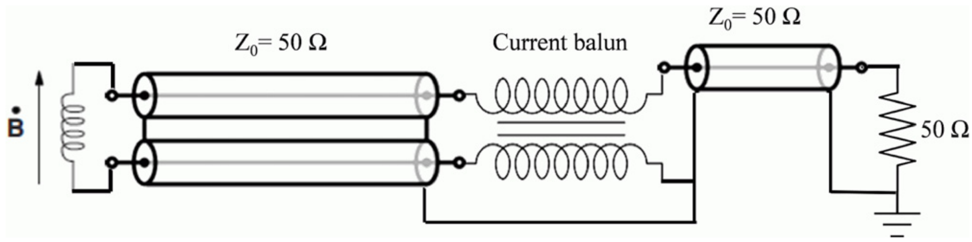

Figure 3. Schematic of the RF magnetic probe used in the MAGPIE. The current balun is used to reject electrostatic signals arising from time-varying plasma potentials.

The procedure to test and select an appropriate scheme for reducing the electrostatic pick up of the probe was based on [26]. Various hybrid combiners and current baluns were tested with the probe inside a Faraday cup with voltages up to 90 V from 3 to 30 MHz. The current balun selected was observed to be capable of reducing the electrostatic coupling between the probe and the Faraday cup at 13.56 MHz by a factor of 50 to 90.

3.4. Optical spectroscopic imaging system

A Doppler coherence imaging camera measures the spatial variation of the ion distribution function. The system is a spatial-heterodyne polarization interferometer as described in [19]. The camera records the bright argon ion emission at 488 nm, against a blackened view dump, along a radial line of sight, and can be translated along the plasma to study the effects of the magnetic field inhomogeneity on the ion distribution function. Some representative brightness profiles are presented below, and a full description of the system and Doppler imaging results will be given in a later paper.

4. Initial characterization of the device

In this section we provide initial measurements of ion density, electron temperature and helicon wave field structures in MAGPIE when operating under the following conditions: 0.41 Pa (3.1 mTorr) in argon, 1.5 ms pulse width, 1.5% duty cycle, and RF power levels of 2.1 and 0.6 kW at 13.56 MHz (table 1). The magnetic field configuration used for this study is shown dotted in figure 6.

During 2.1 kW operations we observed the formation of a plasma with a bright blue core [4, 9, 14, 27, 28] associated with increased ionization and intensity of Ar II ions in the 400 nm range. The 'blue core' is formed above a threshold RF power of 0.8 kW. Using the RF compensated Langmuir probe located at 0.17 m from the source/diffusion chamber interface (in the target region—see figure 1) we measured the following plasma parameters at the centre of the discharge.

The ion density and electron temperature radial profiles for blue-core plasmas are shown in figure 4(*a*) (RF power 2.1 kW). It is observed that both the electron density and temperature are a maximum at the axis and decrease towards the edge. Density in excess of $10^{19}\,\mathrm{m}^{-3}$ can be achieved in argon at higher fill pressures and RF power levels. For higher power short pulses at 7 MHz (20 kW, 1 ms), higher density is obtained in hydrogen and argon plasma. Figure 4(*b*) shows an axial scan of ion density using a symmetric double Langmuir probe. Hydrogen requires higher power for the same density as argon, and increases in plasma density in the magnetic field gradient region are observed in both gases.

Table 1. Typical plasma parameters—see the text about double probe measurements in H_2.

	RF power (kW)	Ion density (m^{-3})	Electron temp. (eV)
Argon:	0.6	7.3×10^{17}	4.2
Argon Blue core	2.1	5.4×10^{18}	4.2
Hydrogen	20	8×10^{18}	5

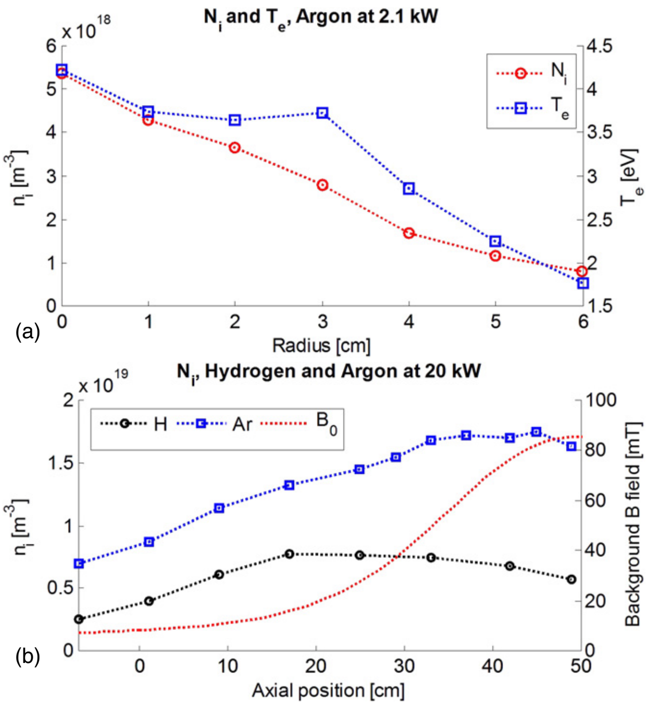

Figure 4. (*a*) Radial profiles of density and temperature for argon plasma during blue-core mode at 0.17 m downstream into the target region during 2.1 kW operation and mirror ratio of 12.5. (*b*) Axial profile of hydrogen and argon plasma density for 20 kW 1 ms pulses, showing density increasing in the magnetic field gradient region. A double probe was used in (*b*) to avoid excessive currents at high density.

A composite image of Ar^+ 488 nm emission as a function of axial position across the mirror field region is shown in figure 5, including brightness profiles at various maximum mirror magnetic field strengths. Compared with figure 4,

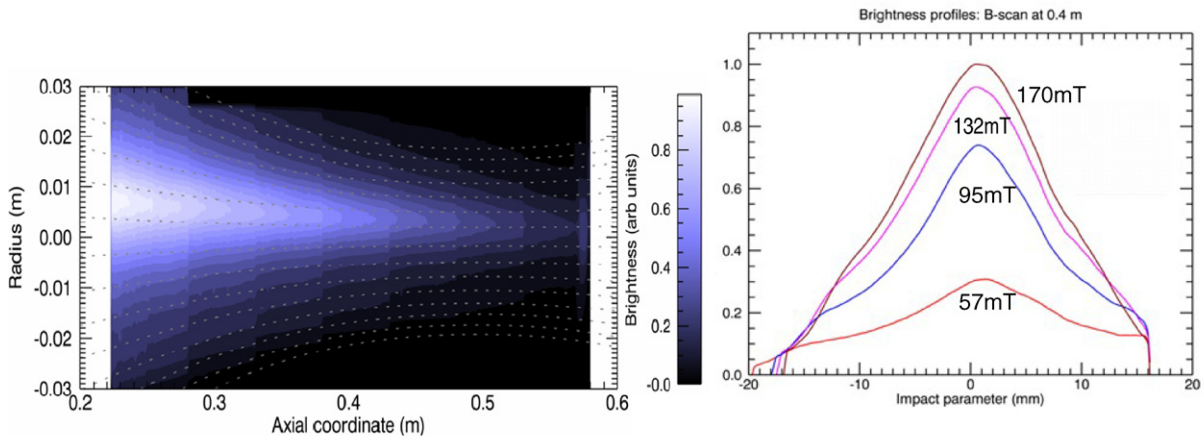

Figure 5. Radial profiles of the line-integrated 488 nm argon ion emission as a function of (*a*) axial position, with superimposed magnetic field lines (dotted, mirror field 85 mT, mirror ratio 12.5) and (*b*) magnetic field strength in the mirror region (z = 0.4 m) during operation at 2.1 kW. The lowest field data (57 mT) correspond to the onset of the centrally peaked emission in the 'blue-core' mode.

the profiles are considerably narrower due to the magnetic compression towards the target region and the dependence on density, clearly illustrated in the composite image. The radial profiles are qualitatively consistent with the measured density and electron temperature profiles assuming the coronal radiation dependence $n_e^2 T_e^{0.5}$ reported in [29]. The increase in brightness with magnetic field strength is consistent with the behaviour of helicon heating which tends to keep the ratio n_e/B_0 constant. Ion Doppler data indicate an approximately rigid body azimuthal flow velocity profile with speeds up to $1000\,\mathrm{m\,s^{-1}}$. Ion temperatures are <1 eV, and preliminary measurements indicate complex axial flows near the ion sound speed in the magnetic field gradient region.

5. Helicon wave activity

Figure 6 shows the axial helicon wave field profiles for non-blue and blue-core plasmas measured downstream of the antenna, the edge of which is located at $z = -3$ cm; additionally, the non-uniform background magnetic field is shown for reference (dotted line). We first note that the exponentially decaying, approximately linearly polarized near field of the antenna on the non-blue-core plasmas (0.6 kW) is observable up to 10 cm downstream of the source/diffusion chamber interface, whereas for the blue-core plasma (2.1 kW) the amplitude of the near field effect is much smaller in comparison with the circularly polarized ($m = +1$) component, which accounts for a minimum of 80% of the total field away from the near field, depending on position. This is supported by the comparison with the measured b_R component of the antenna vacuum field (lower curve, figure 6(*b*)), measured in the absence of plasma with the same antenna current (and correspondingly lower input power). It should be noted that the probe was observed to affect the plasma formation considerably when near the antenna. In particular, the blue-core mode was inaccessible when the magnetic probe approached within 10 cm of the antenna. Thus the observed maximum in the wave amplitude 10 cm downstream from the antenna is likely to be an artefact of the probe preventing transitions to blue-core mode.

As the RF power is increased the attenuation length of the wave amplitude is observed to decrease. The $1/e$ attenuation lengths are 0.35 m and 0.22 m for the non-blue and blue-core plasmas, respectively. The increase attenuation rate of the blue-core mode ($4.5\,\mathrm{m^{-1}}$) implies increased damping and power deposition from the wave to the plasma. For all cases we also observe the b_Z component to be considerably lower than the b_R component, which is consistent with an $m = +1$ helicon mode.

The axial phase variation of b_r allows us to estimate the axial wavelength and phase velocity of the travelling helicon. During non-blue-core operation the wavelength near the antenna, just outside the near field region is comparable to the antenna length, and becomes smaller than the antenna length in the blue-core mode. Furthermore, we observe that at high powers (2.1 kW) the slope of the phase shift increases with axial position, indicating a decrease in phase velocity, whereas at low powers (0.6 kW) the opposite is true.

Both the increasing attenuation rate and the decreasing phase velocity of the b_r component during 2.1 kW plasma operation indicate that there is an increased wave–plasma interaction in the blue-core mode. We note that the simple picture of helicon dispersion ($|k|k_\parallel = q_e\mu_0\omega(n_e/B_0)$) predicts a strong dependence of $k_\parallel$ on both n_e/B_0, but in order to model this device with the considerable gradients in both quantities which are comparable to wavenumbers, we would expect that a full-wave solution would be required.

6. Conclusions and outlook

In this paper we have described MAGPIE, a new linear plasma device constructed to study basic plasma phenomena, test materials under extreme conditions and serve as a test bed for developing diagnostics for the edge region of fusion reactors. We have presented some initial representative results for achieved plasma densities, electron temperatures, argon ion brightness profiles and helicon wave activity observed in the device.

We have observed the $m = +1$ helicon wave field structures downstream of the antenna during non-blue-core and

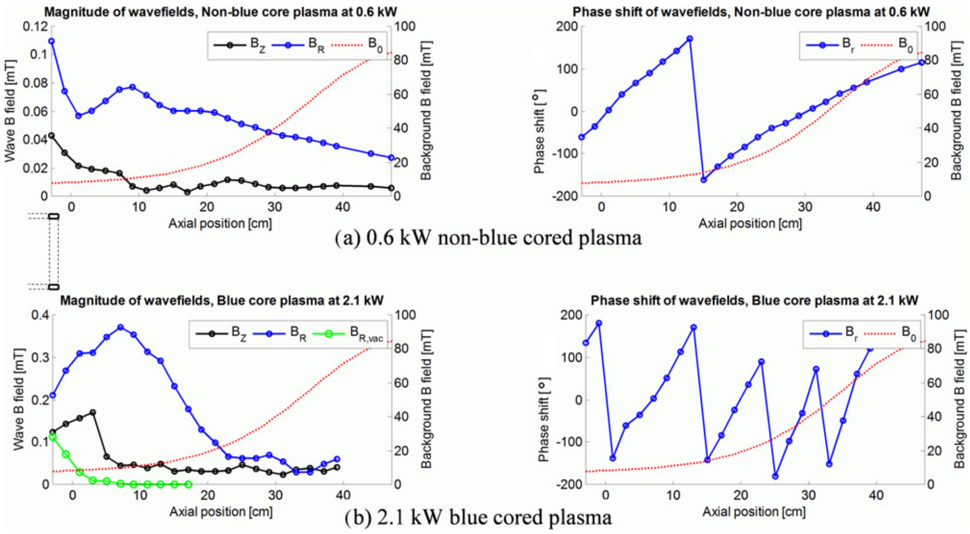

Figure 6. Axial wavefield measurements of helicon waves during (*a*) non-blue-core mode at 0.6 kW and (*b*) blue-core mode at 2.1 kW. The magnitude of the background magnetic field is shown for reference (dotted line). The axial position scale is as for figure 1, and the position of the nearest end of the antenna is indicated.

blue-core operation. We have shown that blue-core plasmas result in a decrease in the axial wavelength, attenuation length and phase velocity of the helicon wave as the wave propagates into the magnetic mirror region. A more detailed helicon wave activity paper will be presented in a separate publication.

MAGPIE is ultimately intended to study plasma–material interactions and test diagnostics relevant to hydrogen/deuterium fusion plasmas. Following some basic physics studies in argon, we aim to optimize the plasma density in hydrogen. This will require higher RF powers which will be available from the two 40 kW RF transmitters and 12 MW DC power supplies of the H-1NF stellarator at the Australian Plasma Fusion Research Facility.

Acknowledgments

The authors are grateful to R Goulding (Oak Ridge National Labs) for advice on design. The Facility was partly funded by the Australian Government under the Super Science Initiative, financed from the Education Investment Fund, and the Future Fellowship Scheme of the Australian Research Grants Council.

References

[1] FESAC 2007 Priorities, gaps and opportunities: towards a long-range strategic plan for magnetic fusion energy DOE/SC-0102, *FESAC Report*

[2] Goebel D M, Campbell G and Conn R W 1984 Plasma surface interaction experimental facility (PISCES) for materials and edge physics studies *J. Nucl. Mater.* **121** 277–82

[3] van Eck H J N *et al* 2007 Pre-design of Magnum-PSI: a new plasma–wall interaction experiment *Fusion Eng. Des.* **82** 1878–83

[4] Boswell R W 1984 Very efficient plasma generation by whistler waves near the lower hybrid frequency *Plasma Phys. Control. Fusion* **26** 1147–62

[5] Degeling A, Sheridan T and Boswell R 1999 Model for relaxation oscillations in a helicon discharge *Phys. Plasmas* **6** 1641

[6] Schröder C, Grulke O, Klinger T and Naulin V 2004 Spatial mode structures of electrostatic drift waves in a collisional cylindrical helicon plasma *Phys. Plasmas* **11** 4249

[7] Schröder C, Grulke O, Klinger T and Naulin V 2005 Drift waves in a high-density cylindrical helicon discharge *Phys. Plasmas* **12** 042103

[8] Light M, Chen F F and Colestock P L 2001 Low frequency electrostatic instability in a helicon plasma *Phys. Plasmas* **8** 4675

[9] Corr C S and Boswell R W 2007 High-beta plasma effects in a low-pressure helicon plasma *Phys. Plasmas* **14** 122503

[10] Corr C S and Boswell R W 2009 Nonlinear instability dynamics in a high-density, high-beta plasma *Phys. Plasmas* **16** 022308

[11] Chang L, Hole M J and Corr C S 2011 A flowing plasma model to describe drift waves in a cylindrical helicon discharge *Phys. Plasmas* **18** 042106

[12] Winglee R, Ziemba T, Giersch L, Prager J, Carscadden J and Roberson B R 2007 Simulation and laboratory validation of magnetic nozzle effects for the high power helicon thruster *Phys. Plasmas* **14** 063501

[13] Charles C 2009 Plasmas for spacecraft propulsion *J. Phys. D: Appl. Phys.* **42** 163001

[14] Guo X M, Scharer J, Mouzouris Y and Louis L 1999 Helicon experiments and simulations in nonuniform magnetic field configurations *Phys. Plasmas* **6** 3400

[15] Prager J, Ziemba T, Winglee R and Roberson B R 2010 Wave propagation downstream of a high power helicon in a dipolelike magnetic field *Phys. Plasmas* **17** 013504

[16] Kikuchi H, Sakawa Y and Shoji T 2002 Hydrogen plasma production in the lower-hybrid frequency range *J. Plasma Fusion Res. Ser.* **5** 302–5

[17] Mori Y, Nakashima H, Baity F W, Goulding R H, Carter M D and Sparks D O 2004 High density hydrogen helicon plasma in a non-uniform magnetic field *Plasma Sources Sci. Technol.* **13** 424–5

[18] Howard J, Michael C, Glass F and Danielsson A 2003 Time-resolved two-dimensional plasma spectroscopy using coherence-imaging techniques *Plasma Phys. Control. Fusion* **45** 1143–66

[19] Howard J 2010 Coherence imaging spectro-polarimetry for magnetic fusion diagnostics *J. Phys. B: At. Mol. Opt. Phys.* **43** 144010

[20] Light M and Chen F F 1995 Helicon wave excitation with helical antennas *Phys. Plasmas* **2** 1084

[21] Mori Y, Nakashima H, Baity F W, Goulding R H, Carter M D and Sparks D O 2006 Focusing magnetic field contribution for helicon plasma on Mini-RFTF *Thin Solid Films* **506–507** 583–7

[22] Cantin A and Gagne R R J 1977 Pressure dependence of electron temperature using rf-floated electrostatic probes in rf plasmas *Appl. Phys. Lett.* **30** 316–9

[23] Swift J D and Schwar M J R 1970 *Electrical Probes for Plasma Diagnostics* (London: Iliffe Books)

[24] Brockhaus A, Borchardt C and Engemann J 1994 Langmuir probe measurements in commercial plasma plants *Plasma Sources Sci. Technol.* **3** 539–44

[25] Cherrington B E 1982 The use of electrostatic probes for plasma diagnostics—a review *Plasma Chem. Plasma Process.* **2** 113–40

[26] Franck C M, Grulke O and Klinger T 2002 Magnetic fluctuation probe design and capacitive pickup rejection *Rev. Sci. Instrum.* **73** 3768–71

[27] Sudit I D and Chen F F 1996 Discharge equilibrium of a helicon plasma *Plasma Sources Sci. Technol.* **5** 43–53

[28] Ellingboe A R, Boswell R W, Booth J P and Sadeghi N 1995 Electron beam pulses produced by helicon-wave excitation *Phys. Plasmas* **2** 1807

[29] Scime E, Hardin R, Biloiu C, Keesee A M and Sun X 2007 Flow, flow shear, and related profiles in helicon plasmas *Phys. Plasmas* **14** 043505